Counting System Complexity and Mental Load Scoring Framework

Status: Draft v2 — pending review
Reference game: 6D S17 DOA DAS LS RSA peek A or T, 75% penetration
Author: Eddie Merkel / BJW research project
Last revised: 2026-04-03

---

1. Purpose

This document defines an objective, reproducible method for scoring the cognitive complexity and mental load of card counting systems. The goal is to produce a scalar score and a flag profile for each system that together answer two questions:

1. How mentally demanding is this system overall?
2. What specifically makes it demanding?

The framework is designed so that every value in the index is either a countable fact from the system definition, an analytical derivation from known parameters, or a measurement from simulation data. No arbitrary weights or subjective judgments are introduced, with one explicitly documented exception described in Section 6.

This framework is intended to replace purely anecdotal judgments of count-system difficulty with a more explicit and reproducible scoring method. It should still be understood as a modeling framework, not a discovered law of cognition.

---

2. Scope

This is a **reference-condition complexity score**, not a universal burden estimate. It measures the cognitive demands of operating a counting system under a fixed reference game configuration. It does not attempt to model every possible game pace, table occupancy, or training environment.

Real mental burden does change with context — heads-up vs full table, pitch vs shoe, deep vs shallow penetration. The reference game controls for those differences. Context modifiers are outside the scope of this framework.

---

3. Reference Game

All simulation-derived components must be measured under identical conditions. The reference game is:

- **Decks:** 6
- **Rules:** S17, DOA, DAS, LS, RSA, peek A or T
- **Penetration:** 75%
- **Rounds:** minimum 1 billion (per BJW evidence threshold)

Comparison across systems is only valid when all systems are simulated under this identical configuration.

---

4. Two Sub-Scores

The index captures two distinct dimensions of cognitive burden:

Sub-score 1 — Instantaneous Load
Mental work required per card and per decision during active play.

Sub-score 2 — Sustained Load
Persistent cognitive state maintained throughout the shoe, independent of any single card or decision.

Each sub-score is computed from its component values. The two sub-scores combine into a single final score as described in Section 6.

---

5. Components

5.1 Component Definitions

ID	Name	Sub-score	Measurement method
C1	Distinct non-zero tag values	Instantaneous	Count from system definition
C2	Tag value range	Instantaneous	Calculated: max tag − min tag
C3	Non-standard tag resolution	Instantaneous	Count of card ranks with fractional or suit-aware tags

ID	Name	Sub-score	Measurement method
C4	TC conversion required	Instantaneous	Binary: 1 if system requires true count conversion, 0 otherwise
C5	Divisor changes per shoe	Instantaneous	Calculated from deck count, precision level, and penetration
C6	RC range at 99th percentile	Sustained	Simulation: width of 1st–99th percentile RC band from count_histogram
C7	Side count severity	**Both**	Tiered 0/1/2: see Section 5.2
C8	Starting count magnitude	Sustained	Analytically derived from system definition and deck count
C9	Proportion of shoe in negative territory	Sustained	Simulation: proportion of rounds with RC < 0 from count_histogram
C10	Key count applies	Sustained	Binary: 1 if system has a key count (unbalanced systems only)
C11	Pivot applies	Sustained	Binary: 1 if system has a pivot point (unbalanced systems only)

5.2 Component Measurement Rules

C1 — Distinct non-zero tag values
Count the number of different non-zero tag values the system assigns to card ranks. A system that assigns only +1 and −1 scores 2. A system that assigns +1, +2, and −2 scores 3.

C2 — Tag value range
`C2 = max_tag - min_tag`

Example: Hi-Lo range = +1 − (−1) = 2. Zen Count range = +2 − (−2) = 4. This measures the spread of the value space the player must operate in.

C3 — Non-standard tag resolution
Count card ranks where the tag value cannot be determined from rank alone. Fractional tags (e.g., Wong Halves: 2=+0.5, 3=+1, 5=+1.5, 7=+0.5) and suit-aware tags (e.g., Red Seven: red 7=+1, black 7=0) both qualify. Score is the count of affected ranks, not the count of individual cards.

C4 — TC conversion required
Binary. Score 1 if the system requires true count conversion for its betting and playing decisions. Score 0 if the system is operated as a running count system. Note: a balanced

system operated without TC conversion is an unusual and suboptimal use case — document via variant declaration if applicable.

C5 — Divisor changes per shoe
The number of times the TC divisor changes during a shoe, a function of deck count, precision level, and penetration. Formula:

> divisor_changes = floor(decks_in_play / precision_unit)

where decks_in_play = deck_count × penetration, and precision_unit = 1.0 (whole deck), 0.5 (half deck), or 0.25 (quarter deck).

For the reference game (6D, 75% penetration):

Precision	Divisor changes per shoe
Whole deck	4
Half deck	8
Quarter deck	18

If C4 = 0 (no TC conversion required), C5 = 0.

C6 — RC range at 99th percentile
Measured from simulation count_histogram at sample_time = bet_decision. Compute the width of the band from the 1st to the 99th percentile of the RC distribution. This measures the working memory envelope the player must operate within during normal play, excluding rare extreme counts that inflate theoretical range calculations.

C7 — Side count severity
Tiered score reflecting both the presence and type of side count:

- `0` = no side count
- `1` = side count used mainly for betting or insurance (e.g., ace side count in Hi-Opt II, Omega II, Uston APC)
- `2` = side count used for playing decisions or multi-parameter index adjustments

C7 contributes to both sub-scores. **This is an intentional weighting choice.** Side counts add both parallel memory load (instantaneous) and persistent reconciliation overhead (sustained). Double-counting C7 reflects that dual burden. This is a modeling decision, not a discovered law.

C8 — Starting count magnitude
The absolute value of the initial running count (IRC) at the start of each shoe. For balanced

systems starting at 0, C8 = 0. For unbalanced systems, IRC varies by deck count and is defined in the system specification. Example: KO 6D IRC = −20, so C8 = 20.

Derived analytically from the system's rules file — no simulation required.

C9 — Proportion of shoe in negative territory
Measured from simulation count_histogram at sample_time = bet_decision. Proportion of rounds where RC < 0. For a balanced system starting at 0 this will be approximately 0.42–0.50. For a deeply negative-starting unbalanced system this will be substantially higher.

C10 — Key count applies
Binary. Score 1 if the system uses a key count threshold that triggers betting or playing decisions. Primarily applies to unbalanced systems (KO, REKO, UBZ 2). Score 0 for balanced systems.

C11 — Pivot applies
Binary. Score 1 if the system uses a pivot point for playing decisions. Primarily applies to unbalanced systems. Score 0 for balanced systems.

---

6. Normalization and Scoring

6.1 Normalization

Because components are measured on different scales, each component is normalized against the maximum value observed for that component across all systems in the study:

> normalized_value(component, system) = raw_value(component, system) / max_raw_value(component, across all systems)

This produces a value in [0, 1] for each component. Binary components (C4, C10, C11) and the C7 tier naturally produce values in [0, 1] after normalization.

Important: The normalization denominator for each component is empirically determined from the study dataset. Adding a new system that sets a new maximum for any component will change the normalized scores for all other systems on that component. This is expected behavior and must be documented when the system set changes.

6.2 Scoring

Sub-score 1 (Instantaneous) = sum of normalized values for C1, C2, C3, C4, C5, C7

Sub-score 2 (Sustained) = sum of normalized values for C6, C7, C8, C9, C10, C11

Final score = mean of all 12 normalized component values:

> final_score = (C1n + C2n + C3n + C4n + C5n + C7n + C6n + C7n + C8n + C9n + C10n + C11n) / 12

C7 appears twice — once per sub-score — because it genuinely taxes both dimensions. This is documented as an explicit weighting choice in Section 5.2.

6.3 Raw Score

Both the raw total and the normalized score are published for each system:

- **raw_complexity_total** — sum of raw (unnormalized) component values. Stable across studies; supports comparison over time even if the system set changes.
- **normalized_complexity_score** — the final score above. Valid for ranking within a fixed study set.

---

7. Flag Profile

A flag is set for any component that scores non-zero for a given system. No threshold, no judgment — if the raw value is non-zero, the flag is set.

Flags provide qualitative fingerprint information that the scalar score alone cannot convey. Two systems with identical final scores can have completely different flag profiles, reflecting different cognitive characters.

---

8. Variant Declaration

A complexity score is valid only for the exact operational variant scored. The same system name can exist in multiple materially different forms — different IRC conventions, with or without side counts, different index packages, different TC divisor conventions.

Every scored system must declare:

- **Tag set used**
- **Side count:** included or excluded, and type if included
- **IRC convention:** starting count value at deck count used
- **TC convention:** balanced with TC, or unbalanced RC-only
- **Deviation package basis:** none, limited, full, or side-count-adjusted

> Scores must not be generalized across alternate IRC, side-count, or index-package conventions unless those conventions are explicitly held constant across the comparison.

---

9. Deviation Disclosure

The base complexity score measures the burden of maintaining and converting the count. It does not score the complexity of a deviation package.

For practical comparisons, each scored system also carries a deviation disclosure:

- `deviations: none` — no index plays used
- `deviations: limited package` — a small set of plays (e.g., insurance only, or a handful of high-value plays)
- `deviations: full package` — a complete index play package equivalent to Illustrious 18 style
- `deviations: side-count-adjusted package` — index plays that incorporate side count information, meaningfully harder than a standard package

> Comparing count systems without their usual deviation package understates their practical ceiling. The base complexity score should be read as the burden of maintaining and converting the count, not the full ceiling of the completed system in use.

Note: a standard full index package (Illustrious 18 style) is roughly equivalent across systems and does not materially distort comparisons. Side-count-adjusted packages are not equivalent and are meaningfully harder — this distinction is why the disclosure tier exists.

---

10. Example Scoring

The following examples use known systems scored against the reference game (6D, 75% penetration). Simulation-required values for Hi-Lo and KO are marked TBD — pending 1B-round runs under the reference game config. Ace-Five simulation values are confirmed from the reference run (1B rounds, seed 12345, March 2026).

10.1 Variant Declarations

Ace-Five

- Tag set: A = −1, 5 = +1, all others = 0
- Side count: none

- IRC: 0
- TC convention: RC only, no TC conversion
- Deviation package: none

Hi-Lo

- Tag set: 2–6 = +1, 7–9 = 0, T/A = −1
- Side count: none
- IRC: 0
- TC convention: balanced, TC required
- Deviation package: full (Illustrious 18 basis) — assumed for scoring; confirm against BJW rules file

KO (6D)

- Tag set: 2–7 = +1, 8–9 = 0, T/A = −1
- Side count: none
- IRC: −20 (6D) — confirm against BJW rules file before publishing
- TC convention: unbalanced, RC only
- Deviation package: limited (Preferred 16 / key count basis) — assumed; confirm

10.2 Raw Component Values

Component	Ace-Five	Hi-Lo	KO (6D)
C1: Distinct tag values	2	2	2
C2: Tag range (max − min)	2	2	2
C3: Non-standard tags	0	0	0
C4: TC required	0	1	0
C5: Divisor changes/shoe	0	8 [1]	0
C6: RC range 99th pct	14 [2]	TBD	TBD
C7: Side count severity	0	0	0
C8: Starting count magnitude	0	0	20 [3]
C9: Prop. shoe RC < 0	0.418 [2]	TBD	TBD
C10: Key count	0	0	1
C11: Pivot	0	0	1

Notes:
[1] Hi-Lo at half-deck precision, 6D 75% pen = 8 divisor changes. Verify precision convention against BJW rules file.
[2] Confirmed from BJW simulation: 1B rounds, 6D S17 DOA DAS RSA LS, 75% pen, seed 12345, March 2026.
[3] KO 6D IRC = −20 per published system specification. Verify against BJW rules file before publishing.

10.3 Normalization Denominators (Current)

Normalization denominators cannot be finalized until the full system dataset is available. Current observable maximums from the three systems scored so far:

Component	Current max	Source
C1	2	All three systems
C2	2	All three systems
C3	0	None yet — needs fractional/suit-aware systems
C4	1	Hi-Lo
C5	8	Hi-Lo (half-deck)
C6	14	Ace-Five (others TBD)
C7	0	None yet — needs system with side count
C8	20	KO 6D
C9	0.418	Ace-Five (others TBD)
C10	1	KO
C11	1	KO

These denominators will shift as higher-complexity systems are added (Wong Halves → C3; Revere RAPC → C1, C2; Hi-Opt II with ace side count → C7). All scores must be recomputed when denominators change.

10.4 Normalized Scores (Illustrative — Incomplete Dataset)

The following normalized scores use current observed maximums only. They are illustrative and will change as the dataset grows.

Component	Ace-Five	Hi-Lo	KO
C1n	1.000	1.000	1.000

Component	Ace-Five	Hi-Lo	KO
C2n	1.000	1.000	1.000
C3n	0.000	0.000	0.000
C4n	0.000	1.000	0.000
C5n	0.000	1.000	0.000
C6n	1.000	TBD	TBD
C7n (×2)	0.000	0.000	0.000
C8n	0.000	0.000	1.000
C9n	1.000	TBD	TBD
C10n	0.000	0.000	1.000
C11n	0.000	0.000	1.000
Final score	TBD [4]	TBD	TBD

[4] Ace-Five final score computable once Hi-Lo and KO simulation runs complete, since those may reset the C6 and C9 normalization denominators.

10.5 Flag Profiles

System	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11
Ace-Five	✓	✓				✓			✓		
Hi-Lo	✓	✓		✓	✓	TBD			TBD		
KO	✓	✓				TBD		✓	TBD	✓	✓

Even with TBD simulation values, the flag profiles already show a meaningful difference: Hi-Lo's complexity is concentrated in TC arithmetic (C4, C5), while KO's comes from negative territory management and threshold tracking (C8, C10, C11). These are different cognitive characters that could produce the same final score.

10.6 Example Output Shape

A complete scored entry for Ace-Five would look like:

```
System: Ace-Five
Variant: no TC, RC only, no side count, IRC=0, no deviations

raw_complexity_total: 4.418
normalized_complexity_score: TBD (pending full dataset)
```

```
side_count_tier: 0
deviations: none

Flags: C1, C2, C6, C9
```

---

11. Simulation Requirements

All systems in the study require 1B-round reference game runs to produce simulation-derived component values (C6, C9). All runs must use the reference game configuration in Section 3. Result files must include count_histogram at sample_time = bet_decision.

Simulation runs for the complexity index are independent of other BJW research runs and can be commissioned and executed in parallel.

11.1 Candidate System List

The candidate systems are drawn from the qfit.com card counting strategy reference (https://www.qfit.com/card-counting.htm). A system is included in the study if an optimized bet spread can be sourced — either from published literature or from CVCX. Systems without a sourceable optimized bet spread are excluded.

System	Type	Level	Bet spread source	Include
Canfield Expert	B1	1	TBD	TBD
Canfield Master	B2	2	TBD	TBD
Hi-Lo	B1	1	TBD	TBD
Hi-Opt I	B1	1	TBD	TBD
Hi-Opt II	B2	2	TBD	TBD
KISS 2	US1	1	TBD	TBD
KISS 3	US1	1	TBD	TBD
K-O	UC1	1	TBD	TBD
Mentor	B2	2	TBD	TBD
Omega II	B2	2	TBD	TBD
Red Seven	USC1	1	TBD	TBD
REKO	UC1	1	TBD	TBD
Revere Adv. Plus Minus	B1	1	TBD	TBD
Revere Point Count	B2	2	TBD	TBD

System	Type	Level	Bet spread source	Include
Revere RAPC	B4	4	TBD	TBD
Revere 14 Count	B4	4	TBD	TBD
Silver Fox	B1	1	TBD	TBD
UBZ 2	U2	2	TBD	TBD
Uston Adv. Plus Minus	B1	1	TBD	TBD
Uston APC	B3	3	TBD	TBD
Uston SS	U3	3	TBD	TBD
Wong Halves	B3	3	TBD	TBD
Zen Count	B2	2	TBD	TBD

Type key (from qfit.com): B=balanced, U=unbalanced, S=suit-aware, C=compromise indexes, number=level.

11.2 Notes on Specific Systems

KISS 2 and KISS 3 — The 2-card is tagged 0/1 (suit-aware). This affects C3. Verify against the KISS system specification before finalizing C3 values.

Red Seven — The 7-card is tagged 0/1 by suit (red 7 = +1, black 7 = 0). C3 = 1. Confirm against BJW rules file.

Wong Halves — Uses fractional tags (0.5, 1.0, 1.5). C3 will be non-zero. Will set normalization ceiling for C3 across the study.

Revere RAPC and Revere 14 Count — Level 4 systems. Will likely set normalization ceilings for C1 and C2, compressing all other systems' scores on those components. Expected behavior.

Hi-Opt II, Omega II, Uston APC — Commonly used with ace side counts. Variant declaration must specify whether the side count is included. If included, C7 = 1 (betting/insurance tier). This changes the normalized score for all systems on the C7 dimension and must be decided consistently across the study before simulations are run.

---

12. Known Limitations and Open Questions

1. **C3 not yet stress-tested.** Wong Halves, KISS 2, KISS 3, and Red Seven will be the first systems to produce non-zero C3 values. The definition of "affected ranks" should be verified

against those systems before treating C3 as settled.

2. **C5 precision convention.** The document assumes the stated precision level for each system. If a system is commonly practiced at a different precision than its published convention (e.g., Hi-Lo practiced at full-deck rather than half-deck), the variant declaration must capture this.
3. **Normalization denominator stability.** The dataset is currently small. All scores must be recomputed when denominators change. The raw_complexity_total is the stable artifact for tracking a system's absolute profile over time.
4. **C7 = 2 systems not yet in dataset.** No system with a side-count-adjusted playing index is yet scored. When one is added, the C7 normalization ceiling will shift from 1 to 2 (or to 1 if normalized against the tier scale directly — this needs a decision before the first C7=2 system is added).
5. **KO IRC value.** Stated as −20 for 6D based on published convention. Must be verified against the BJW KO rules file before the example scoring is published.

---

13. Document History

Date	Change
2026-03-31	v1 — initial draft
2026-04-03	v2 — C2 formula corrected; C7 double-weight disclosed as explicit modeling choice; raw score added alongside normalized; C7 upgraded from binary to 0/1/2 severity tier; deviation disclosure layer added; variant declaration requirement added; scope limit added; C5 kept unchanged; context modifiers explicitly excluded